SHREYA VENKATESH

shreyavenkatesh9@gmail.com ● LinkedIn ● Google Scholar

EDUCATION

2026	Ph.D., Mechanical Engineering	University of Colorado Boulder
2023	M.S., Mechanical Engineering	University of Colorado Boulder
2021	B.S., Biomedical Engineering	Cornell University

RESEARCH PUBLICATIONS

2026 — (In preparation) **Venkatesh S**, Ferguson VL, Lynch ME. “Advanced glycation end-product accumulation in the onco-aging environment as a driver of bone metastatic breast cancer.” *Review*.

2026 — (In preparation) **Venkatesh S**, Thompson WR, Lynch ME. “Mechanically loaded breast cancer cells prevent sclerostin downregulation by mature osteocytes in a co-culture model of the metastatic vicious cycle.”

2026 — **Venkatesh S**, Groves AR, Joshi P (co-first authors). “Addressing Elderly Loneliness with Artificial Intelligence Companionship.” *MIT Science Policy Review, Volume VII*.

2026 — **Venkatesh S**, et al. “Radiation modulates the mechanoresponse of bone-homing triple-negative breast cancer cells.” *Tissue Engineering Part A*, *Mechanobiology in Tissue Engineering Special Issue*.

2025 — **Venkatesh S**, et al. “High-fidelity computational fluid dynamics modeling to simulate perfusion through a bone-mimicking scaffold.” *Computers in Biology and Medicine*.

2024 — Lang BJ, Holton KM, Guerrero-Gimenez ME, Okusha Y, Magahis PT, Shi A, Neguse M, **Venkatesh S**, et al. “Heat shock protein 72 supports extracellular matrix production in metastatic mammary tumors.” *Cell Stress and Chaperones*.

OTHER PUBLICATIONS & GRANT-WRITING EXPERIENCE

2026 — **Venkatesh S**, Groves AR, Joshi P (co-first authors). “Addressing Elderly Loneliness with Artificial Intelligence Companionship.” *MIT Science Policy Review, Volume VII*.

2026 — **Venkatesh S**, Affsprung D, Mattson A. “The Past Is Our Present: A History of America's Complicated Relationship with Science.” *Science Policy in a SNAP.*

2025 — **National Institute of Aging (NIA) R36 Dissertation Awards**, PAR-24-052. “Aging-induced AGE accumulation heightens bone fragility in metastasis by dysregulating osteocytes.” Total: $140,000. Impact Score: 30. (grant decisions suspended)

2023 — Hayes A, Sundaram V, Williams G, **Venkatesh S**, et al. “Meeting Schools Where They Are: Integrating Engineering Outreach Curriculum in the Classroom Without Forcing an Agenda.” *American Society for Engineering Education: Rocky Mountain Section Conference*.

EXPERIENCE

Jun – Aug 2026 — **Colorado Science & Technology Policy Program Summer Scholar**
Institute for Science & Policy — *Denver Museum of Nature & Science*

- Synthesized multidisciplinary scientific literature and stakeholder engagement into policy memos for Colorado state legislators as part of the multi-sector plan on aging (topics included social isolation, AI use by older adults, etc.).
- Researched and analyzed policy issues covering multiple fields, including water, transportation, renewable energy, education, and Medicaid.

Aug 2021 – 2026 — **Cancer and Bone Doctoral Researcher**
PI: Dr. Maureen E. Lynch — *University of Colorado Boulder*

- Integrated in vitro experiments and computational estimates of internal mechanics to investigate the skeleton's biological response to exercise and disease.
- Demonstrated proficiency at distilling complex scientific information into compelling takeaways for expert and lay audiences (3-Minute Thesis Finalist, Best Poster/Oral Presentation Awards, Young Investigator Travel Awards, talks at the Denver Museum of Natural History).

Jan 2019 – May 2021 **Tendon Mechanobiology Research Assistant**
PI: Dr. Nelly Andarawis-Puri *Cornell University*

- Characterized the role of poorly understood non-collagenous proteins in tendon fatigue damage accumulation via mouse models of overuse injuries.
- Designed a physical therapy glove for at-home rehabilitation of the hand and developed a mobile app paired with the device to relay information between patient and physician/therapist.

Sept 2018 – Dec 2019 **Immunotherapy & Cell Engineering Research Assistant**
PI: Dr. Ankur Singh *Cornell University*

- Engineered microfluidic devices and embedded lymphoma cells in hydrogels to validate the use of 3D human organ culture models for translational cancer research.

May – Aug 2018 **Radiation Oncology Research Intern**
Mentors: Dr. Benjamin Lang, Dr. Stuart Calderwood *Beth Israel Deaconess Medical Center*

- Established data analysis pipelines for gene and protein expression analysis in R.
- Quantified expression of a tumor biomarker (Hsp70) within different subtypes of human breast cancer and examined molecular pathways activated within irradiated tumor cells.

PRESENTATIONS

Oral Presentations

2026	Three Minute Thesis Finalist Talk. "The Infamous Cancer Cell Hatches a New Evil Plan…to Destroy Bone!"
2025	**Venkatesh S***, et al. "SERPINE1 Underpins the Tumor-Osteocyte Response to Fluid Shear and Irradiation." *ASBMR Annual Meeting*. Podium. [awarded]
2025	**Venkatesh S***, et al. "Characterizing the biochemical responses that upset bone homeostasis in the metastatic vicious cycle." *Advances in Mineral Metabolism (AIMM) ASBMR John Haddad Young Investigator Meeting*. Late-Breaking Talk.
2025	Session Moderator (Co-Chair). "Oncology Meets Osteology: Advances in Cancer & Bone Research." *AIMM ASBMR John Haddad Young Investigator Meeting*.
2024	**Venkatesh S***, et al. "Dynamic Flow Alters the Breast Cancer Cell Mechanoresponse in a 3D Perfusion Model of Bone Metastasis." *MechBio Symposium*. Podium. [awarded]
2023	Hayes A*, Sundaram V, Williams G, **Venkatesh S***, et al. "Meeting Schools Where They Are: Integrating Engineering Outreach Curriculum in the Classroom Without Forcing an Agenda." *American Society for Engineering Education: RMS Conference*. Podium.
2019	"And in This Corner: Hsp70 and Breast Cancer." *Cornell Undergraduate Research Board, CURBx Talks*. Public talks in the style of TEDx. [awarded]

Poster Presentations

2026	**Venkatesh S***, et al. "SERPINE1 is Overlooked in the Tumor-Osteocyte Response to Fluid Shear and Irradiation." *AAAS Annual Meeting*. [awarded]
2026	**Venkatesh S***, et al. "SERPINE1 Underpins the Tumor-Osteocyte Response to Fluid Shear and Irradiation." *Bones and Teeth Gordon Research Conference*.
2025	**Venkatesh S***, et al. "Dual Role of Periostin in Mediating the Bone Mechanoresponse in Metastasis In Vivo." *ASBMR Annual Meeting*.

2025	"Characterizing the Vicious Cycle of Bone Destruction by Breast Cancer Metastasis." *Achievement Rewards for College Scientists (ARCS) Colorado Spring Gala*.
2023	**Venkatesh S***, et al. "Dynamic Fluid Flow Alters Breast Cancer Cell Mechanoresponse in a 3D Perfusion Model of Bone Metastasis." *Cancer and Bone Society Meeting*.

*Presenting author.

SELECTED HONORS & AWARDS

2026	First Place. Technology, Engineering & Math Category, *AAAS E-Poster Competition*.
2026	Three Minute Thesis (3MT) Finalist.
2025	Young Investigator Travel Grant, *ASBMR Annual Meeting*.
2024 – 2026	Graduate Research Scholar, *ARCS Foundation*.
2024	Best Graduate Scientific Brief & Poster Presentation, *MechBio Symposium*.
2024	Committee for Equity in Mechanical Engineering Doctoral Service Award, *University of Colorado Boulder*.
2022 – 2023	Community Engagement & Outreach Awards, *University of Colorado Boulder*.
2023	AAAS CASE Workshop Sponsorship (one of two *Boulder* students).
2023	Future K-12 Leaders Service Award, *University of Colorado Boulder*.
2023	Visiting Research Scholar, *University of Massachusetts Amherst*.
2022	GRFP Honorable Mention, *National Science Foundation*.
2021	Chair's Graduate Fellowship, *University of Colorado Boulder*.
2021	Best Science Communication and Applied Research Finalist, *CURB Spring Symposium*.
2019	People's Choice Award, *'CURBx' Talks*.
2017	John McMullen Dean's Scholar, *Cornell University*.

TEACHING, OUTREACH, AND SERVICE

2025 – Current **District Visit Leader & Team Member**, *Scientist Network for Advancing Policy (SNAP)*

- Organized district office visits for 3 states (AZ, CO, MI). Advocated for maintaining FY26 federal science funding at or above the levels of previous fiscal year budgets.
- Maintained relationships with staffers at Senator Hickenlooper's (D-CO) office. Tracked legislative activity related to AI and emerging technologies. Synthesized complex technical information into concise, actionable summaries.

2022 – 2026	**Court-Appointed Special Advocate**, *Boulder County*
2021 – 2026	**Outreach Program Lead**, *University of Colorado Boulder*
2022 – 2024	**Graduate Student Representative**, *University of Colorado Boulder*
2021 – 2023	**Project-Based Learning Mentor**, *University of Colorado Boulder*
2018 – 2021	**Children's Science Program Volunteer**, *Ithaca Sciencenter*

TECHNICAL SKILLS

Software:

MATLAB. R. Arduino. Java. Python. Microsoft Excel. Excel Solver. Autodesk Fusion 360. SolidWorks. APM Planner. QGroundControl. Cura. OpenSim. CircAdapt. SnapGene. GraphPad. ImageJ. Image Studio Lite. ANSYS. ABAQUS. Simpleware ScanIP. OpenFOAM. DragonFly. IVIS Lumina. Supercomputing Clusters. Bash (Unix shell command-line interpreter).

Mechanical and Electrical:

Mechanical Device Design. Wiring. CAD. 3D Printing. Laser Cutting. Soldering. Circuit Design. Oscilloscope. PDMS Microfluidic Device Fabrication. Tensile Testing. Dynamic/Fatigue Loading. Multiaxial Ball Burst Testing. Perfusion & Compression Bioreactor Systems.

Chemical and Biological:

Cell & Tissue Culture. Cell Cryopreservation. Western Blot. Real-Time qPCR. Histology. Flow Cytometry. Organoid Generation. Boyden Chamber Assays. RNA Purification. Restriction Enzyme Digests. Gel Electrophoresis. Bacterial Transformation & Transfection. BCA & Ninhydrin-Based Assays. ELISA. Immunohistochemistry. Hydrogel & Scaffold Fabrication. Next-Generation Sequencing. Ex Vivo Cultures. Raman Spectroscopy.

Other Technical Skills:

Digital Image Correlation (DIC). Mouse Husbandry. Modern Regression Analysis. Nonparametric Statistical Methods. Fluorescence Microscopy. 2D, 3D Microscopy. Spectrophotometry. Second Harmonic Generation (SHG) Imaging. Basic App Development.

MEDIA

2026	"Meet 3MT Finalist Shreya Venkatesh." University of Colorado Boulder, Graduate School.
2025	"Fight for continuation of scientific funding." *Scottsdale Progress*, Letters to the Editor. McClintock Letters Initiative, *SNAP Coalition*.
2023	"Researchers at CU Boulder land NSF Award for air and soil quality education."
2023	"CU Boulder students get high schoolers excited about STEM fields through outreach event."